Task(250 points): Substitution Cipher Decryption using File Handling

- **Note:1** This task involves understanding basic file handling and character manipulation in C. You are provided with a ciphertext file that was encrypted using a Substitution Cipher mechanism. Your goal is to decrypt it back to the original message using the provided letter mapping.
- **Note:2** The substituion mapping given in the table in the below question is just an example. Every student should take two input files, an input file (cipher(your roll number).txt) from cipher text folder and also the other input file (mapping(your of roll number).txt) from mapping text folder in your department folders. This mapping is used as substitution table for your cipher text input file.
- **Note:3** For this assignment you will not be given any expected output file to verify your outputis correct or not. You have to make sure your self that you are obtaining a valid decryption i.e decryption is resulting some meaning full words or not.
- **Note:4** At 7:45PM a google form will be shared which require you to paste your output. Deadline for filling google form will be sharp 8:00PM.
- **Note:5** You will be given expected output folder at 10:00PM.
- **Note:6** As part of tutorial you have to update your code if it is not giving expected output. Later you have to upload your code along with tutorial codes.
- **Note:7** Using internet before 7:45PM is considered as malpractice.

Problem Statement

Many years ago, people did not have computers or the internet to send secret messages. They used simple tricks to hide their messages so that only trusted people could read them. One such old method is called the **Substitution Cipher**. In this method, every letter in a message is replaced by another letter of the alphabet. For example, ‘A’ may become ‘Q’, ‘B’ may become ‘W’, and so on for all 26 letters. This way, the message looks strange and cannot be read easily by others. Only the person who knows the secret mapping can change it back to the real message. This type of secret writing was used by soldiers and kings to send safe messages. Today, you will work as a small “code breaker” to read one such hidden message. A file named `ciphertext.txt` contains a secret message written using substitution cipher rules. Each letter in the file was changed based on a fixed mapping that you already know. Your task is to write a C program to open that file and read the encrypted message. Then, using the substitution mapping table, you should replace every letter with its real letter. If the letter in the file is ‘Q’, and according to the table ‘Q’ means ‘A’, then you must write ‘A’. Spaces in the message should not be changed and must stay in the same place. After decoding all the letters, you must write the final, readable message into another file called `output.txt`. This will be the original plain message that was hidden earlier. Use file handling functions like `fopen()`, `fgetc()`, and `fputc()` to perform all file operations. Make sure your program checks if the file can be opened before reading or writing. At the end, you will have successfully broken the code and revealed the secret message!

Cipher Mechanism

Each alphabet letter (A–Z) has been substituted according to the following mapping table:

Plain Letter	**Cipher Letter**	**Plain Letter**	**Cipher Letter**
A	Q	N	F
B	W	O	G
C	E	P	H
D	R	Q	J
E	T	R	K
F	Y	S	L
G	U	T	Z
H	I	U	X
I	O	V	C
J	P	W	V
K	A	X	B
L	S	Y	N
M	D	Z	M

Requirements

1. The program should:
   - Open and read the file `ciphertext.txt`.
   - Decrypt each character based on the substitution mapping given above.
   - Preserve spaces (they are not encrypted).
   - Write the decrypted plaintext to `output.txt`.
2. Use only uppercase English letters (A–Z) for simplicity.
3. Handle file opening errors properly (e.g., print `Error: cannot open file.`).1

Example

Input file 1(`ciphertext.txt`):

```
ITSSG VGKSR
```

Input file 2(`mapping.txt`):

```
refer above table
```

Output file (`output.txt`):

```
HELLO WORLD
```

Explanation:
Using the given substitution table:

$$I \to H, \quad T \to E, \quad S \to L, \quad G \to O, \quad V \to W, \quad K \to R$$

Thus, the ciphertext `ITSSG VGKSR` decrypts to `HELLO WORLD`.